

Munkres 12

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November 2025

1 Homology of Groups

Let F be a group and $N \triangleleft F$. Denote by $q : F \rightarrow F/N$, $p : F \rightarrow F/[F, F]$ the projection maps. Then $q(F)/[q(F), q(F)] \cong p(F)/p(N)$.

Proof: Consider the composition

$$F \rightarrow F/N \rightarrow (F/N)/[F/N, F/N].$$

Since $(F/N)/[F/N, F/N]$ is Abelian, this map descends to the epimorphism

$$\varphi : F/[F, F] \rightarrow (F/N)/[F/N, F/N].$$

We compute the kernel.

Suppose $f[F, F] \in \ker \phi$. Then equivalently $(fN) \in [F/N, F/N]$. Since $N \triangleleft F$, we equivalently have $fN = gN$ for some $g \in [F, F]$, or $f \in gN = Ng$, or $f[F, F] = n[F, F]$ for some $n \in N$, or $f[F, F] \in p(N)$.

Let $\varphi : G \rightarrow H$ be an epimorphism. Let $N \triangleleft G$ be normal. Then $\varphi(N) \triangleleft H$.

Proof: Since $N \neq \emptyset$, $\varphi(N) \neq \emptyset$. Suppose $\varphi(n), \varphi(m) \in \varphi(N)$ for $n, m \in N$. Then by homomorphism properties $\varphi(n)\varphi(m) = \varphi(nm) \in \varphi(N)$, since N is closed under $\cdot$.

Suppose $\varphi(n) \in N$. Then $\varphi(n)^{-1} = \varphi(n^{-1}) \in \varphi(N)$, since N is closed under inverses. It follows that $\varphi(N) < H$.

Suppose $h \in H$. Since φ is surjective, let $g \in G$ such that $\varphi(g) = h$. Then $h\varphi(N)h^{-1} = \varphi(g)\varphi(N)\varphi(g^{-1}) = \varphi(gNg^{-1}) = \varphi(N)$, since $gNg^{-1} = N$. It follows $\varphi(N) \triangleleft H$.

2 Computation of Nonabelian Fundamental Groups

The fundamental group of the n -fold torus for $n > 1$ is nonabelian.

Proof: Let X be the n -fold torus. $\pi_1(X) = \langle a_1, b_1, \dots, a_n, b_n \mid [a_1, b_1] \cdots [a_n, b_n] \rangle$. By the extension property of free groups, there exists a homomorphism $\varphi : F_{2n} \rightarrow Q_8$ such that $\varphi(a_1) = \varphi(a_2) = i$, $\varphi(b_1) = \varphi(b_2) = j$, and $\varphi(a_i) = \varphi(b_i) = 1$ for $i > 2$. Since $\langle i, j \rangle = Q_8$, φ is surjective. Observe that

$$((ij)(ji)^{-1})^2 = (k(-k)^{-1})^2 = (k^2)^2 = k^4 = 1.$$

Hence, φ descends to the epimorphism $\bar{\varphi} : \pi_1(X) \rightarrow Q_8$. Since Q_8 is nonabelian, so is $\pi_1(X)$.

The fundamental group of the m -fold projective plane for $m > 1$ is nonabelian.

Let X be the m -fold projective plane. By the extension property of free groups, there exists a homomorphism $\varphi : F_m \rightarrow \mathbb{Z}_2 * \mathbb{Z}_2$ given by $\varphi(a_1) = 1_1$, $\varphi(a_2) = 1_2$ and $\varphi(a_i) = 0$ for $i > 2$. Since $\mathbb{Z}_2 * \mathbb{Z}_2 = \langle 1_1, 1_2 \rangle$, φ is surjective. Observe that $1_1 1_1 1_2 1_1 = 0$, so φ descends to the epimorphism $\bar{\varphi} : \pi_1(X) \rightarrow \mathbb{Z}_2 * \mathbb{Z}_2$. Since $\mathbb{Z}_2 * \mathbb{Z}_2$ is nonabelian, so is $\pi_1(X)$.